WEEK – 1

SESSION – 1

GIT COMMANDS

git init

git status

git add file.txt

git commit -m "Initial commit"

git branch feature1

git checkout feature1

git add folder/

git commit -m "Added folder"

git push origin feature1

SESSION – 2

```
mysql> CREATE DATABASE fullstack;
Query OK, 1 row affected (0.01 sec)

mysql> USE fullstack;
Database changed
mysql> CREATE TABLE Student (
    ->     VTU_Number VARCHAR(20) PRIMARY KEY,
    ->     Name VARCHAR(100),
    ->     Email VARCHAR(100),
    ->     Phone VARCHAR(15),
    ->     Department VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql> CREATE TABLE Course (
    ->     Course_Code VARCHAR(10) PRIMARY KEY,
    ->     Course_Name VARCHAR(100),
    ->     Faculty_Id INT,
    ->     Student_ID VARCHAR(20),
    ->     Faculty_Email VARCHAR(100),
    ->     FOREIGN KEY (Student_ID) REFERENCES Student(VTU_Number)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Student VALUES
    -> ('1MS23CS001', 'Arjun Kumar', 'arjun@mail.com', '9845011223', 'CSE'),
    -> ('1MS23CS002', 'Sneha Rao', 'sneha@mail.com', '9845022334', 'IT'),
    -> ('1MS23CS003', 'Rahul Nair', 'rahul@mail.com', '9845033445', 'CSE'),
    -> ('1MS23CS004', 'Priya Das', 'priya@mail.com', '9845044556', 'ECE'),
    -> ('1MS23CS005', 'Amit Singh', 'amit@mail.com', '9845055667', 'IT');
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Course VALUES
    -> ('CS101', 'Database Management', 101, '1MS23CS001', 'prof_smith@vtu.edu'),
    -> ('IS202', 'Data Structures', 102, '1MS23CS002', 'prof_jones@vtu.edu'),
    -> ('CS103', 'Operating Systems', 101, '1MS23CS003', 'prof_smith@vtu.edu'),
    -> ('EC301', 'Digital Electronics', 103, '1MS23CS004', 'prof_kumar@vtu.edu'),
    -> ('IS204', 'Web Programming', 104, '1MS23CS005', 'prof_leila@vtu.edu');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Student WHERE Department = 'CSE';
+------------+-------------+----------------+------------+------------+
| VTU_Number | Name        | Email          | Phone      | Department |
+------------+-------------+----------------+------------+------------+
| 1MS23CS001 | Arjun Kumar | arjun@mail.com | 9845011223 | CSE        |
| 1MS23CS003 | Rahul Nair  | rahul@mail.com | 9845033445 | CSE        |
+------------+-------------+----------------+------------+------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Course ORDER BY Course_Name ASC;
+-------------+---------------------+------------+------------+--------------------
| Course_Code | Course_Name         | Faculty_Id | Student_ID | Faculty_Email
+-------------+---------------------+------------+------------+--------------------
| IS202       | Data Structures     |        102 | 1MS23CS002 | prof_jones@vtu.edu
| CS101       | Database Management |        101 | 1MS23CS001 | prof_smith@vtu.edu
| EC301       | Digital Electronics |        103 | 1MS23CS004 | prof_kumar@vtu.edu
| CS103       | Operating Systems   |        101 | 1MS23CS003 | prof_smith@vtu.edu
| IS204       | Web Programming     |        104 | 1MS23CS005 | prof_leila@vtu.edu
+-------------+---------------------+------------+------------+--------------------
5 rows in set (0.02 sec)

mysql> SELECT Name, Email FROM Student WHERE Email LIKE '%@mail.com';
+-------------+----------------+
| Name        | Email          |
+-------------+----------------+
| Arjun Kumar | arjun@mail.com |
| Sneha Rao   | sneha@mail.com |
| Rahul Nair  | rahul@mail.com |
| Priya Das   | priya@mail.com |
| Amit Singh  | amit@mail.com  |
+-------------+----------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Student LIMIT 3;
+------------+-------------+----------------+------------+------------+
| VTU_Number | Name        | Email          | Phone      | Department |
+------------+-------------+----------------+------------+------------+
| 1MS23CS001 | Arjun Kumar | arjun@mail.com | 9845011223 | CSE        |
| 1MS23CS002 | Sneha Rao   | sneha@mail.com | 9845022334 | IT         |
| 1MS23CS003 | Rahul Nair  | rahul@mail.com | 9845033445 | CSE        |
+------------+-------------+----------------+------------+------------+
3 rows in set (0.00 sec)
```

SESSION – 3

```
mysql> SELECT * FROM Student;
+------------+-------------+----------------+------------+------------+
| VTU_Number | Name        | Email          | Phone      | Department |
+------------+-------------+----------------+------------+------------+
| 1MS23CS001 | Arjun Kumar | arjun@mail.com | 9845011223 | CSE        |
| 1MS23CS002 | Sneha Rao   | sneha@mail.com | 9845022334 | IT         |
| 1MS23CS003 | Rahul Nair  | rahul@mail.com | 9845033445 | CSE        |
| 1MS23CS004 | Priya Das   | priya@mail.com | 9845044556 | ECE        |
| 1MS23CS005 | Amit Singh  | amit@mail.com  | 9845055667 | IT         |
+------------+-------------+----------------+------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS Total_Students FROM Student;
+----------------+
| Total_Students |
+----------------+
|              5 |
+----------------+
1 row in set (0.00 sec)

mysql> SELECT COUNT(DISTINCT Department) AS Department_Count FROM Student;
+------------------+
| Department_Count |
+------------------+
|                3 |
+------------------+
1 row in set (0.02 sec)

mysql> SELECT MAX(Faculty_Id) AS Highest_Faculty_ID FROM Course;
+--------------------+
| Highest_Faculty_ID |
+--------------------+
|                104 |
+--------------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM Student
    -> ORDER BY VTU_Number ASC;
+------------+-------------+----------------+------------+------------+
| VTU_Number | Name        | Email          | Phone      | Department |
+------------+-------------+----------------+------------+------------+
| 1MS23CS001 | Arjun Kumar | arjun@mail.com | 9845011223 | CSE        |
| 1MS23CS002 | Sneha Rao   | sneha@mail.com | 9845022334 | IT         |
| 1MS23CS003 | Rahul Nair  | rahul@mail.com | 9845033445 | CSE        |
| 1MS23CS004 | Priya Das   | priya@mail.com | 9845044556 | ECE        |
| 1MS23CS005 | Amit Singh  | amit@mail.com  | 9845055667 | IT         |
+------------+-------------+----------------+------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Student
    -> ORDER BY VTU_Number DESC;
+------------+-------------+----------------+------------+------------+
| VTU_Number | Name        | Email          | Phone      | Department |
+------------+-------------+----------------+------------+------------+
| 1MS23CS005 | Amit Singh  | amit@mail.com  | 9845055667 | IT         |
| 1MS23CS004 | Priya Das   | priya@mail.com | 9845044556 | ECE        |
| 1MS23CS003 | Rahul Nair  | rahul@mail.com | 9845033445 | CSE        |
| 1MS23CS002 | Sneha Rao   | sneha@mail.com | 9845022334 | IT         |
| 1MS23CS001 | Arjun Kumar | arjun@mail.com | 9845011223 | CSE        |
+------------+-------------+----------------+------------+------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Student
    -> WHERE Department = 'CSE';
+------------+-------------+----------------+------------+------------+
| VTU_Number | Name        | Email          | Phone      | Department |
+------------+-------------+----------------+------------+------------+
| 1MS23CS001 | Arjun Kumar | arjun@mail.com | 9845011223 | CSE        |
| 1MS23CS003 | Rahul Nair  | rahul@mail.com | 9845033445 | CSE        |
+------------+-------------+----------------+------------+------------+
2 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     s.VTU_Number,
    ->     s.Name AS Student_Name,
    ->     c.Course_Name,
    ->     c.Faculty_Id,
    ->     c.Faculty_Email
    -> FROM Student s
    -> INNER JOIN Course c ON s.VTU_Number = c.Student_ID;
+------------+-------------+--------------------+------------+--------------------
| VTU_Number | Student_Name | Course_Name        | Faculty_Id | Faculty_Email
+------------+-------------+--------------------+------------+--------------------
| 1MS23CS001 | Arjun Kumar  | Database Management |        101 | prof_smith@vtu.edu
| 1MS23CS002 | Sneha Rao    | Data Structures    |        102 | prof_jones@vtu.edu
| 1MS23CS003 | Rahul Nair   | Operating Systems  |        101 | prof_smith@vtu.edu
| 1MS23CS004 | Priya Das    | Digital Electronics |        103 | prof_kumar@vtu.edu
| 1MS23CS005 | Amit Singh   | Web Programming    |        104 | prof_leila@vtu.edu
+------------+-------------+--------------------+------------+--------------------
5 rows in set (0.00 sec)

mysql> SELECT
    ->     s.VTU_Number,
    ->     s.Name,
    ->     c.Course_Name
    -> FROM Student s
    -> LEFT JOIN Course c ON s.VTU_Number = c.Student_ID;
+------------+-------------+--------------------+
| VTU_Number | Name        | Course_Name        |
+------------+-------------+--------------------+
| 1MS23CS001 | Arjun Kumar | Database Management |
| 1MS23CS002 | Sneha Rao   | Data Structures    |
| 1MS23CS003 | Rahul Nair  | Operating Systems  |
| 1MS23CS004 | Priya Das   | Digital Electronics |
| 1MS23CS005 | Amit Singh  | Web Programming    |
+------------+-------------+--------------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT
    ->     s.Name,
    ->     c.Course_Code,
    ->     c.Course_Name
    -> FROM Student s
    -> RIGHT JOIN Course c ON s.VTU_Number = c.Student_ID;
+-------------+-------------+--------------------+
| Name        | Course_Code | Course_Name        |
+-------------+-------------+--------------------+
| Arjun Kumar | CS101       | Database Management |
| Rahul Nair  | CS103       | Operating Systems   |
| Priya Das   | EC301       | Digital Electronics |
| Sneha Rao   | IS202       | Data Structures     |
| Amit Singh  | IS204       | Web Programming     |
+-------------+-------------+--------------------+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT s.Name, c.Course_Name FROM Student s
    -> LEFT JOIN Course c ON s.VTU_Number = c.Student_ID
    -> UNION
    -> SELECT s.Name, c.Course_Name FROM Student s
    -> RIGHT JOIN Course c ON s.VTU_Number = c.Student_ID;
+-------------+--------------------+
| Name        | Course_Name        |
+-------------+--------------------+
| Arjun Kumar | Database Management |
| Sneha Rao   | Data Structures     |
| Rahul Nair  | Operating Systems   |
| Priya Das   | Digital Electronics |
| Amit Singh  | Web Programming     |
+-------------+--------------------+
5 rows in set (0.02 sec)
```

```
mysql>
mysql> SELECT s.Name, c.Course_Name
    -> FROM Student s
    -> CROSS JOIN Course c;
+-------------+--------------------+
| Name        | Course_Name        |
+-------------+--------------------+
| Arjun Kumar | Database Management |
| Sneha Rao   | Database Management |
| Rahul Nair  | Database Management |
| Priya Das   | Database Management |
| Amit Singh  | Database Management |
| Arjun Kumar | Operating Systems   |
| Sneha Rao   | Operating Systems   |
| Rahul Nair  | Operating Systems   |
| Priya Das   | Operating Systems   |
| Amit Singh  | Operating Systems   |
| Arjun Kumar | Digital Electronics |
| Sneha Rao   | Digital Electronics |
| Rahul Nair  | Digital Electronics |
| Priya Das   | Digital Electronics |
| Amit Singh  | Digital Electronics |
| Arjun Kumar | Data Structures     |
| Sneha Rao   | Data Structures     |
| Rahul Nair  | Data Structures     |
| Priya Das   | Data Structures     |
| Amit Singh  | Data Structures     |
| Arjun Kumar | Web Programming     |
| Sneha Rao   | Web Programming     |
| Rahul Nair  | Web Programming     |
| Priya Das   | Web Programming     |
| Amit Singh  | Web Programming     |
+-------------+--------------------+
```

SESSION – 4

```
mysql> SELECT course, COUNT(*) AS total_students
    -> FROM students
    -> GROUP BY course;
ERROR 1146 (42S02): Table 'fullstack.students' doesn't exist
mysql> SELECT Department, COUNT(*) AS Total_Students
    -> FROM Student
    -> GROUP BY Department;
+------------+----------------+
| Department | Total_Students |
+------------+----------------+
| CSE        |              2 |
| ECE        |              1 |
| IT         |              2 |
+------------+----------------+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Faculty_Courses (
    ->     Course_Code VARCHAR(10),
    ->     Course_Name VARCHAR(100),
    ->     Faculty_Id INT,
    ->     Faculty_Email VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> SELECT *
    -> FROM Course
    -> WHERE Faculty_Id = 101;
+-------------+---------------------+------------+------------+--------------------+
| Course_Code | Course_Name         | Faculty_Id | Student_ID | Faculty_Email      |
+-------------+---------------------+------------+------------+--------------------+
| CS101       | Database Management |        101 | 1MS23CS001 | prof_smith@vtu.edu |
| CS103       | Operating Systems   |        101 | 1MS23CS003 | prof_smith@vtu.edu |
+-------------+---------------------+------------+------------+--------------------+
```

```
mysql> INSERT INTO Faculty_Courses (Course_Code, Course_Name, Faculty_Id, Faculty_Email)
    -> SELECT Course_Code, Course_Name, Faculty_Id, Faculty_Email
    -> FROM Course
    -> WHERE Faculty_Id = 101;
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Faculty_Courses;
+-------------+---------------------+------------+--------------------+
| Course_Code | Course_Name         | Faculty_Id | Faculty_Email      |
+-------------+---------------------+------------+--------------------+
| CS101       | Database Management |        101 | prof_smith@vtu.edu |
| CS103       | Operating Systems   |        101 | prof_smith@vtu.edu |
+-------------+---------------------+------------+--------------------+
2 rows in set (0.00 sec)

mysql> SELECT *
    -> FROM Student
    -> WHERE Department = 'CSE'
    -> LIMIT 5;
+------------+-------------+----------------+------------+------------+
| VTU_Number | Name        | Email          | Phone      | Department |
+------------+-------------+----------------+------------+------------+
| 1MS23CS001 | Arjun Kumar | arjun@mail.com | 9845011223 | CSE        |
| 1MS23CS003 | Rahul Nair  | rahul@mail.com | 9845033445 | CSE        |
+------------+-------------+----------------+------------+------------+
```

```
mysql> SELECT CONCAT('+91', Phone) AS New_Phone
    -> FROM Student;
+------------------+
| New_Phone        |
+------------------+
| +91+919845011223 |
| +919845022334    |
| +91+919845033445 |
| +919845044556    |
| +919845055667    |
+------------------+
5 rows in set (0.00 sec)
```

SESSION – 5

```
mysql> BEGIN;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Student
    -> SET Phone = '+911234567890'
    -> WHERE VTU_Number = '1MS23CS001';
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Student
    -> SET Phone = '+916666666666'
    -> WHERE VTU_Number = '1MS23CS001';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE Student
    -> SET Phone = '+915555555555'
    -> WHERE VTU_Number = '1MS23CS002';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE Student
    -> SET Phone = '+914444444444'
    -> WHERE VTU_Number = '1MS23CS003';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO sp1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION AddNumbers(a INT, b INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     RETURN a + b;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT AddNumbers(5, 10);
+-------------------+
| AddNumbers(5, 10) |
+-------------------+
|                15 |
+-------------------+
1 row in set (0.00 sec)
```